

ResTIAuGo: A Cognitive Architecture for Team Robotics

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I. INTRODUCTION

Documentation: milwd.github.io/cogar_ws/

Group H proposed a multi-robot software architecture for a team of TIAGo robots tasked with autonomously operating in a restaurant setting. This report investigates the proposed architecture through the lens of cognitive embodiment, applying the core cognitive functions of perception, attention, action selection, and memory. Rather than being a domain-specific algorithm, a cognitive architecture constitutes a domain-general framework that encodes, acquires, and applies an agent's knowledge to generate flexible, goal-directed behavior, serving as a computational model of a general theory of intelligence. To ground our analysis, we focus on three critical components: Simultaneous Localization and Mapping (SLAM) from the Navigation subsystem, Orchestration and Coordination from the Server subsystem, and Action Reasoning from the Brain subsystem.

This paper draws on several solutions from the literature. LVI-SAM [1] can construct a robust and drift-corrected internal map of the environment. The Motion Planning Diffusion (MPD) [2] leverages diffusion models to learn priors over trajectory distributions. The DIARC architecture [3], developed for natural human-robot interaction, comprises perceptual, central, and action-processing subsystems. Finally, the Multi-Robot Task Allocation using Combinatorial Auctions (MRTACA) framework [4] offers a symbolic approach to dynamic task allocation among heterogeneous agents.

II. PERCEPTION

Perception is a cognitive function responsible for transforming sensor inputs into meaningful representations of the world. This function is distributed across multiple components of the system.

SLAM is a core perceptual component that performs spatial inference by fusing data from multiple sensors. SLAM operates at a mid- to high-level of abstraction, combining heterogeneous sensory inputs—such as vision, proprioception, and spatial cues—into coherent symbolic or semi-symbolic world models like grid maps and object graphs. It handles temporal and spatial disambiguation, often through timestamp alignment and feature matching. In the LVI-SAM [1] architecture, a robust SLAM system is presented that integrates LiDAR, IMU, and camera data. This multimodal fusion compensates for the weaknesses of individual sensors, producing a consistent feature map and drift-free localization. The authors demonstrate how strong inter-frame data association yields more reliable state estimation. However, LVI-SAM processes raw camera data and does not incorporate sonar information, while our

solution relies on high-level camera information and data from sonar for obstacle avoidance.

Perception in the orchestration system functions at a semantic level, interpreting the robots' availability state, task urgency, and spatial constraints. In cognitive terms, this is similar to how a human scans a task list and evaluates it against their current workload and resources. Although our current architecture does not yet implement it, a typical cost function for allocation would consider a robot's position relative to the target table and its operational state. These elements constitute structured perception within the orchestration component, enabling task assignment.

Action Reasoning incorporates the state of manipulators through encoders, positions of nearby objects from SLAM, or the status of navigation paths. In symbolic systems, such information is represented using discrete propositions like `object-at(x, y)` or `gripper-free`. In contrast, emergent systems use probabilistic estimates or learned feature vectors that encode similar knowledge. This component thus functions as a perceptual interpreter: mapping state and task into formats suitable for planning and control. Its value lies in enabling robots to make context-sensitive decisions—such as whether an object can be grasped or if a path is clear—based on inferred knowledge rather than direct sensing.

III. ATTENTION

Attention mechanisms are critical for efficiently managing limited resources by shifting focus to more relevant information. Followed is the assessment of attention in the mentioned core components.

Attention in SLAM is of high importance since it is achieved by restricting our focus to specific features, suppressing irrelevant signals, and selecting key aspects of the environment. Rather than processing all sensory data equally, SLAM prioritizes stable and salient features—such as tables and dishes—based on task relevance or upstream attentional cues from vision or task context. The LVI-SAM [1] system employs key frames selection to focus on video frames with the most reliable information, simultaneously suppressing noisy or unreliable sensor inputs.

At the orchestration level, attention involves limiting consideration to relevant subsets of robots and orders—such as those available, physically near the kitchen, or within operational range. Attentional selection here may also emerge from distributed heuristics or be explicitly coded as priority queues or greedy task-matchers. For example, in multi-robot task allocation (MRTA) frameworks, the auction process implicitly suppresses irrelevant robot-task pairings by focusing only on viable bids. Similarly, in cognitive architectures like DIARC [3], the action manager uses utility functions to suppress less

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important goals, illustrating the role of attention in managing task relevance and resource allocation.

Within action reasoning, attention dynamically filters and prioritizes sensory feedback based on the current task stage, e.g. while grasping, the system focuses primarily on the gripper's position and disregards unrelated base status. Although explicit attentional models are rare, goal conditioning methods, such as those described in the MPD paper [2], implicitly guide focus by constraining planning processes to align with task objectives.

IV. ACTION SELECTION

Action selection handles the decision-making process whereby robots choose the next action to execute based on the current environment, goals, and internal states.

Orchestration plays a central role in action selection by assigning tasks to specific robots and managing dynamic task allocation under changing conditions. This process may be symbolic, employing rule-based schedulers that assign tasks based on logical conditions, or distributed, relying on bidding or auction mechanisms where robots propose bids based on local states. The orchestration system typically employs a centralized Winner-Take-All (WTA) model, where a global controller evaluates all robot states and task demands to select the optimal assignment, ensuring conflict-free coordination, such as preventing a robot from serving a table simultaneously with another. The MRTACA [4] approach, using combinatorial auctions, exemplifies this by searching for the highest-value task allocation through an auctioneer algorithm, making it the core decision-making mechanism for multi-robot task assignment.

Action reasoning is responsible for selecting specific low-level actions required to achieve current task goals based on the robot's state and environmental context. It considers action preconditions, current progress, and task goals to decide whether to grasp an object, move to a location, or perform another operation. This component employs a variety of methods, including symbolic planning approaches like finite state machines (FSM), hierarchical task networks (HTN), or PDDL-based planners, reactive rule-based systems, or hybrid methods that combine symbolic reasoning with learned control policies. For example, the MPD paper [2] introduces an action selection algorithm based on diffusion models, which balances symbolic goal representations with data-driven adaptability.

V. MEMORY

Memory in cognitive architectures supports the storage, retrieval, and updating of information. It is commonly divided into sensory, working, and long-term memory.

The SLAM module exemplifies spatial long-term memory by maintaining a persistent, interpretable map of the environment, refined over time for drift correction and global consistency, as seen in LVI-SAM [1]. Working memory is represented by the active local map used for planning, while sensory memory includes recent sensor inputs awaiting integration. This mirrors DIARC's memory model, where long-term and working memory serve global and local spatial processing, respectively [3].

The orchestration system depends on working memory to track robot states and tasks, often via a shared blackboard. Long-term memory encodes scheduling rules and resource constraints, while episodic memory records past assignments to improve future decisions. MRTACA [4] demonstrates a hybrid memory model storing current bids, robot statuses, and task queues for dynamic task allocation.

Action reasoning uses sensory memory to track task execution, episodic memory for learning from past actions, and semantic memory to enforce general constraints and procedural rules. Hybrid approaches integrate symbolic traces with learned representations for adaptive behavior. MPD [2] exemplifies this by combining episodic trajectories with semantic priors to train adaptable motion planners.

VI. CONCLUSION

This report has examined the multi-robot architecture proposed by Group H through the lens of core cognitive principles. By focusing on three key components—Simultaneous Localization and Mapping, Orchestration and Coordination, and Action Reasoning, we highlighted how each embodies cognitive functionalities essential for autonomous operation in dynamic environments.

The SLAM module serves as a foundational perceptual-memory system, providing a persistent and interpretable spatial representation that supports higher-level reasoning and navigation. The orchestration and coordination layer exemplifies attention and decision-making processes by prioritizing task allocation and managing resource constraints within a multi-robot team. Meanwhile, the action reasoning component integrates policies to adaptively select actions based on context and current state.

While the architecture demonstrates some strengths aligned with cognitive theories, it also reveals areas for improvement, notably in task and progress evaluation and integrated task scheduling. Incorporating mechanisms to monitor overall system performance and dynamically adjust plans would enhance robustness and adaptability.

Comparisons with established solutions like LVI-SAM, DIARC, and MRTACA further validate the relevance of the cognitive-inspired design and suggest promising directions for future enhancements. Embedding cognitive principles within robotic architectures supports intelligent behavior and provides a structured framework to improve complex multi-robot systems.

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